

IAS PRELIMS/MAINS



SCIENCE & TECHNOLOGY



A COMPREHENSIVE COVERAGE

MODULE - 36

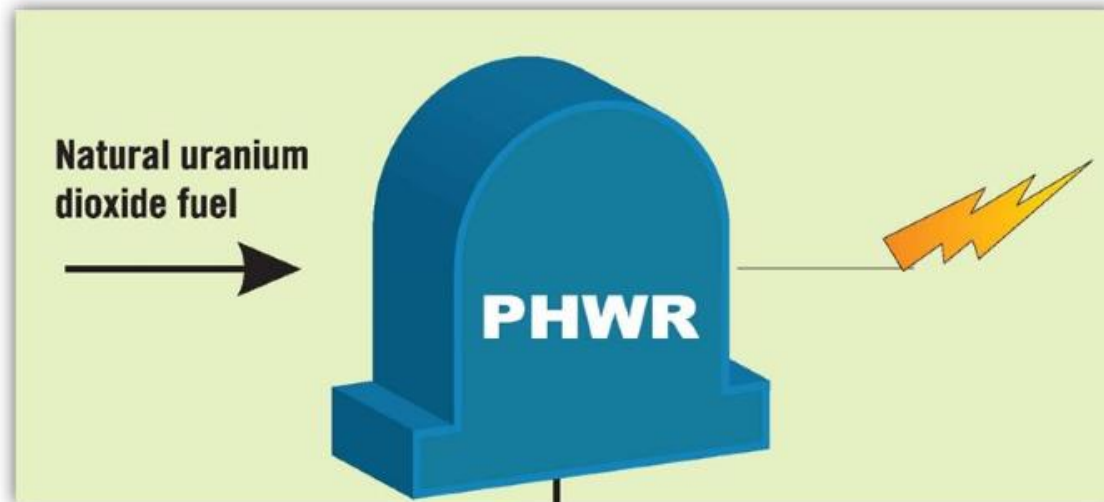
NUCLEAR TECHNOLOGY -2

NUCLEAR TECHNOLOGY

INDIA'S THREE STAGE NUCLEAR PROGRAM

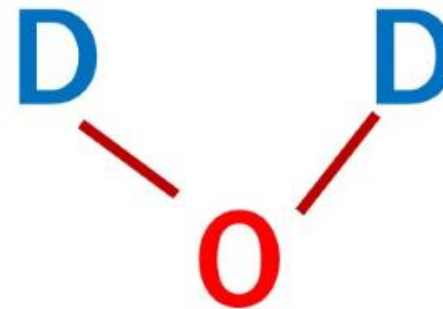
FIRST STAGE

- Reactor Used – Pressurized Heavy Water Reactor
- Fuel Used – Uranium
- Moderator and Coolant – Heavy water and Light water.



HEAVY WATER

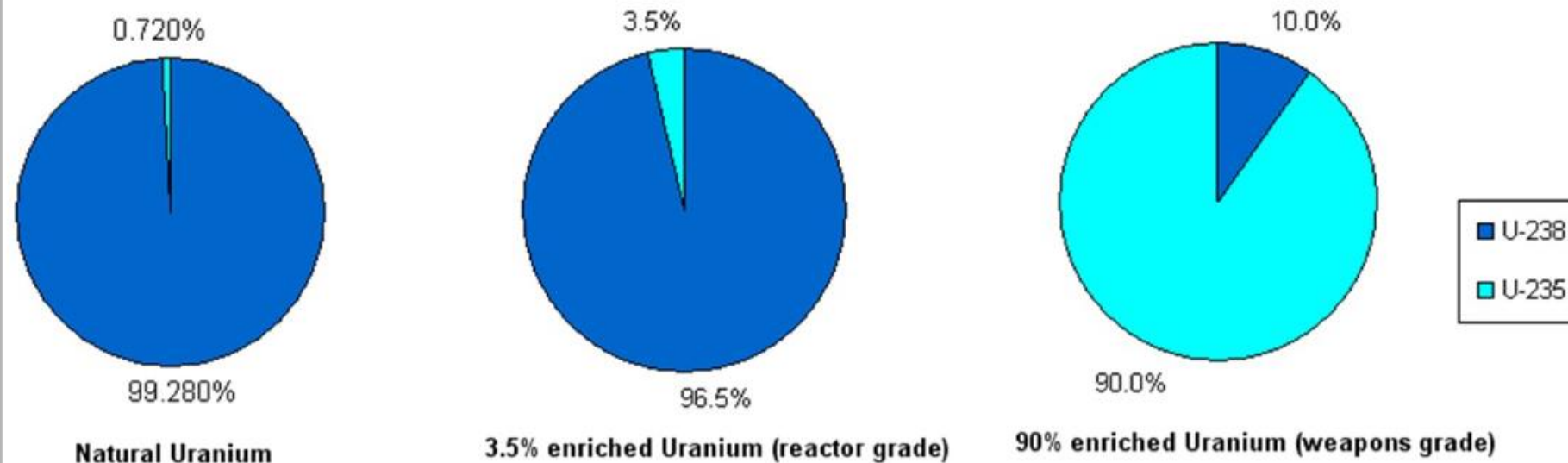
Heavy Water is a form of water that contains a larger than normal amount of Hydrogen isotope Deuterium, rather than the common Hydrogen – 1 isotope that makes up most of the Hydrogen in normal water.



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URANIUM ENRICHMENT

- Uranium in its natural form consists of 0.7% of isotope Uranium - 235 which is fissile. The remaining 99.3% is Uranium - 238 which is fertile but not fissile.
- So, uranium is enriched before put to use, which is increasing the proportion of U - 235 in comparison to U - 238.



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REACTION PROCESS

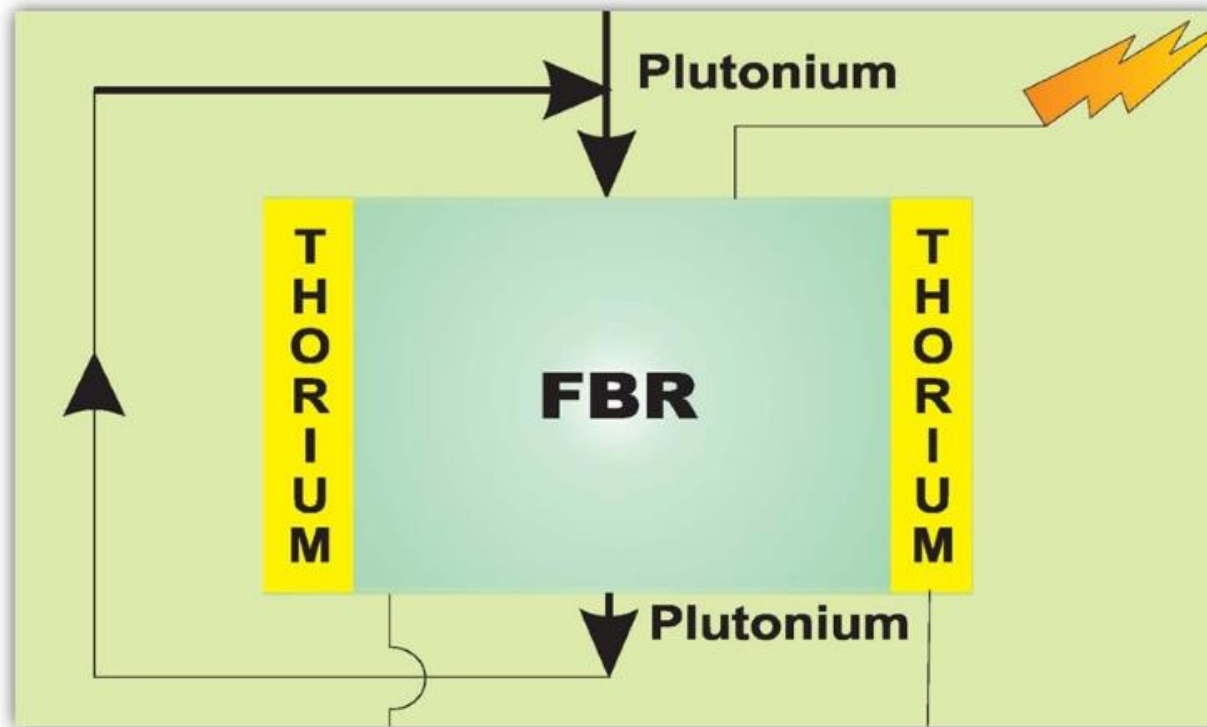
- When the reaction starts, neutrons are bombarded on the nuclei. This splits them in 2 parts or sometimes 3 parts. This releases a significant amount of energy, along with it **3 neutrons are also released which collide with and split the other U - 235 nuclei, this results in a self-sustaining chain reaction.**
- As mentioned, the neutrons released per fission (3) are more than the neutrons consumed per fission (1). So, overtime there are surplus neutrons.
- **Some of these surplus neutrons are used for transmutation of U - 238 into Plutonium - 239, which is fissile.**
- **So, the spent fuel of the first stage consists of Pu -239 and a large amount of U - 238 which didn't get any neutrons to get converted into Pu. This fuel then goes into the 2nd stage.**
- Use of Heavy Water – Heavy Water moderates the speed of neutrons released from the reaction, so that the probability of fission of U - 235 is increased significantly.



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SECOND STAGE

- Reactor Used – Fast Breeder Reactor
- Fuel – Mixture of U – 238 and Pu-239
- No Moderator is needed; Sodium is used as coolant



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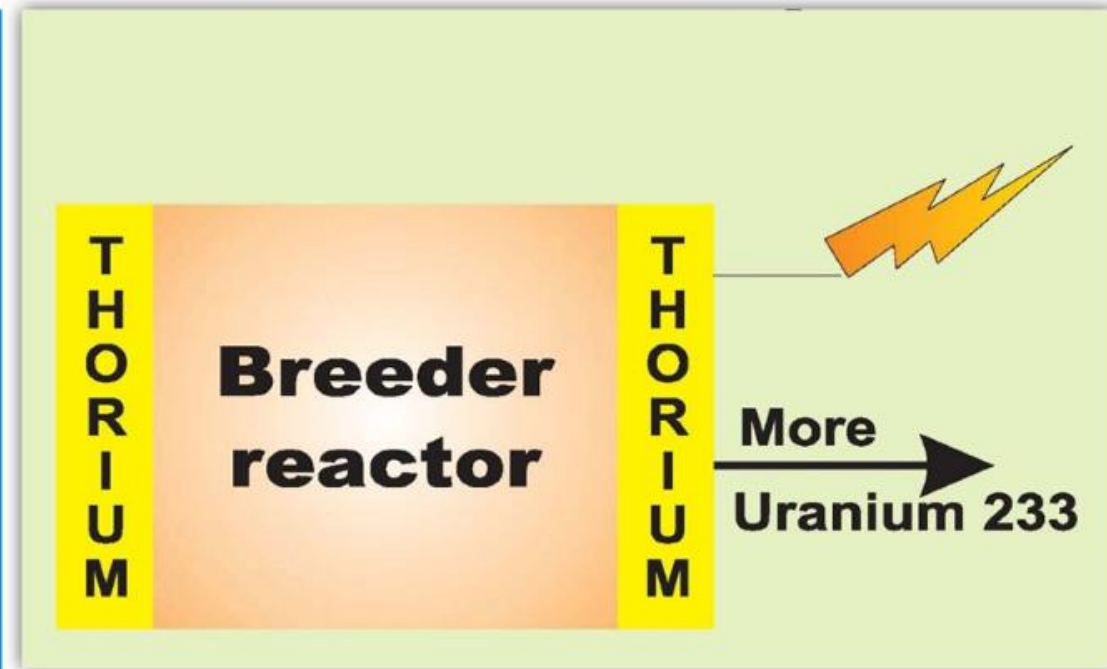
REACTION STAGE

- Pu – 239 is a fissile material and on fission reaction, extra neutrons are produced than the chain reaction needs.
- These extra neutrons are absorbed by U – 238 and converted into Pu – 239. So, in the 2nd stage more fuel is generated than consumed, so it is called 'Breeder Reactor'.
- This reactor doesn't use Moderator as at high speed of neutrons only does Pu – 239 undergo fission and produce extra neutrons.
- Once sufficient power generation capacity is achieved at 2nd stage and there are good reserves of Pu – 239, Th – 232 will be introduced into the reactor as a blanket material which will be transmuted to U -233 (this is done when the fuel is Pu – 239 alone, instead of the mixture of Pu – 239 and U – 238).

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THIRD STAGE

- In this stage, Thorium Based Reactors (Advanced Heavy Water Reactors) are used that can be refuelled using India's large Thorium reserves, which is then converted to U-233 inside the reactor.
- Thorium gives less radioactive waste than Uranium and Plutonium. Also, Plutonium can be separated from waste and can be used to make nuclear bombs, this is not possible with Thorium.
- According to the three-stage programme, Indian nuclear energy could grow to about 10 GW through PHWRs fuelled by domestic uranium, and the growth above that will come from FBRs till about 50GW. The third stage will be deployed only after this capacity has been achieved.



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THIRD STAGE

- So, large scale thorium deployment is only to be expected 3–4 decades after the commercial operation of fast breeder reactors. Full exploitation of India's domestic thorium reserves will not likely occur until after the year 2050.
- 1st stage of the programme is undertaken by Nuclear Power Corporation of India Ltd, whereas 2nd stage is under Bharatiya Nabhikiya Vidyut Nigam Ltd (Bhavini) for FBRs.
- India now has the capacity to build completely indigenous PHWRs. These are supported by two research centers – Bhabha Atomic Research Center and Indira Gandhi Center for Atomic Research.
- Currently there are twenty-two nuclear power reactors with a total installed capacity of 6,780 MW.



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